University of Central Punjab

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FACULTY OF INFORMATION TECHNOLOGY

Programming Fundamentals Section B8 – Spring 2025

LAB MIDTERM EXAM

FILE HANDLING AND FUNCTIONS

April 28, 2025 (Monday)

Version B for Odd Registration Number

Name: ___________________________ Registration Number: ___________________________

Time Allowed: **115** minutes

About Midterm

- *Make Sure to Attempt Correct Version of the Exam, **VERSION A** is for **EVEN REGISTRATION NUMBER** and **VERSION B** is for **ODD REGISTRATION NUMBER**.*
- ***Attempt of incorrect Exam Version is zero marks***
- ***Each Task** has Its **Own Submission Time**, please make sure to Submit on Time or that Task will be Missed.*
- *Make Sure to Save Your Files Frequently.*
- ***Code Must Compile and run for it to be Graded**.*
- *Read the Task Description Very carefully.*
- ***Not Following Good Code Writing may result in up to 100 marks loss***
- *Using any **built in functions, string** will result in **zero marks**.*
- *Describe in **pure English**, in comments, how will you do each of the problems and how will you meet the constraints (**missing this, up to 70 % marks may be lost**)*
- *Evaluation in the last hour*
- ***Sit According to your Registration number in Even/Odd Sequence (Exam will be cancelled for both if both registration number are even or odd**.*
- *More Marks for using more functions than required*
- ***ZERO marks for cin, cout, loops, file handling and No logic in main() other than function calls***

There is **no restriction call by-value, by-reference** etc. for Functions.

Submission Instructions

- Each Task Must be Submitted in a Single.cpp File with Each File Must be named with your registration number and task number e.g., F24_L1S24BSCS0001_midterm_Task1.cpp (**otherwise your work will not be graded)**

Task 1:

Part A

Write a C++ program to manage a **mini contacts manager** using arrays:

- Each contact has:
  - A **name** (character array)
  - A **phone number** (character array)
- The program should allow the user to:
  - **Add a contact** (store name and phone number)
  - **Display all contacts**
  - **Search for a contact by name**
  - **Find the contact with the longest name**

Requirements:

- Use **character arrays** (no `string` class).
- Use **at least four functions**:
  - `addContact()`
  - `displayContacts()`
  - `searchContactByName()`
  - `findLongestNameContact()`

Example Interaction:

Add Contact:

Name: Alice

Phone: 1234567890

Add Contact:

Name: Bob

Phone: 9876543210

Display Contacts:

1. Alice (1234567890)

2. Bob (9876543210)

Search Contact:

Enter name to search: Alice

Phone Number: 1234567890

Longest Name Contact:

```
Alice (1234567890)
```

Part B

Write a C++ program that:

- Reads a **4-digit number** from a file named **"input.txt"** using a **character array**.
- **Validates** that the number contains **at least two even digits** (use a function `validateEvenDigits()`).
    - o If not, display `"Invalid number: Not enough even digits"` and stop.
- Separates each digit and stores them into **four different integer variables** using a function **`extractDigits()`**.
- Forms **two new numbers**:
    - o One by arranging the digits in **reverse order**.
    - o One by arranging only **even digits first and odd digits after** (in the original order they appear) using a function `rearrangeEvenOdd()`.
- Displays both new numbers.
- Calculates and displays the **difference** between these two numbers.
- If the difference is **an odd number**, display `"Odd Difference"`, otherwise display `"Even Difference"`.

Requirements:

- Use **file handling** to read the number.
- Use a **character array** to store the number.
- Use **integer arrays** to help rearrange digits.
- Create at least **5 functions**:

    - `readNumberFromFile()`
    - `validateEvenDigits()`
    - `extractDigits()`
    - `rearrangeEvenOdd()`
    - `displayResults()`
- Keep **main()** minimal: **only function calls** allowed.

File content ("input.txt"):

5732

Program Sample Output:

Original number: 5732

Number after reversing digits: 2375

Number after placing even digits first: 2735

Difference between the two numbers: 160

Even Difference

Task 2:

Part A

Objective:
Write a C++ program that:

1. Takes a 7-digit integer as input.
2. Stores each digit in an integer array.
3. Reverses the digits and stores them in another array.
4. Prints the reversed number.
5. Saves the original and reversed numbers to a CSV file.

Instructions:

- **Input:** A 7-digit integer from the user.
- **Processing:**
    - Store digits in an array.
    - Reverse the digits and store them in another array.
    - Save both original and reversed numbers to a CSV file.
- **Output:**

  - Display the reversed number.
  - Save the original and reversed numbers in "numbers.csv".

Sample Input:

Enter a number (maximum 7 digits): 1234567

Sample Output:

```
Reversed Digits: 7 6 5 4 3 2 1
```

CSV File Output (saved in "numbers.csv"):

```
Original, Reversed
1234567, 7654321
```

Part B

Write a C++ program that:

- Reads a sentence (maximum 300 characters) from the user.
- Finds all words starting with 'c' or 'C'.
- Counts:
  - The number of such words.
  - The total number of characters in those words.
- If no such word is found, display an appropriate message.

Save the results into a file `"cwords.txt"`.

Requirements:

- Use **character arrays** for processing.
- Implement **at least three functions**:
  - One to find and count words.
  - One to count characters.
  - One to save results to a file.

Sample Input:

Can cats climb cold cliffs carefully?

Sample Output:

Words starting with 'c' or 'C': 6

Total characters: 29

If No 'C' Words:

No word starting with 'c' found.